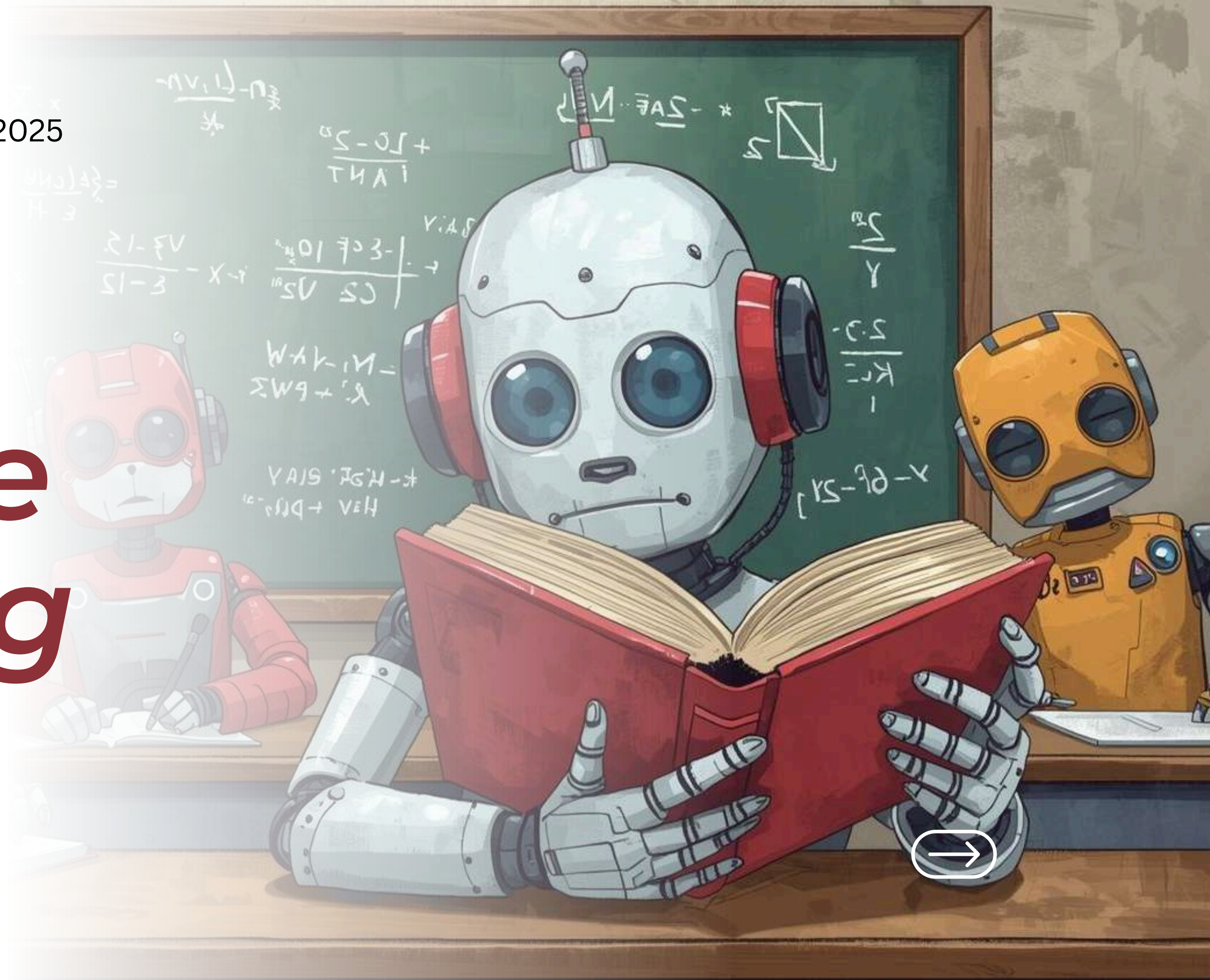


Sun , 21 Sept 2025
Bootcamp of Mathematics (BOOM) 2025

Intro to *Machine Learning*

Muhammad Nanda S. / Data Scientist



Objective

By the end of this session you will:

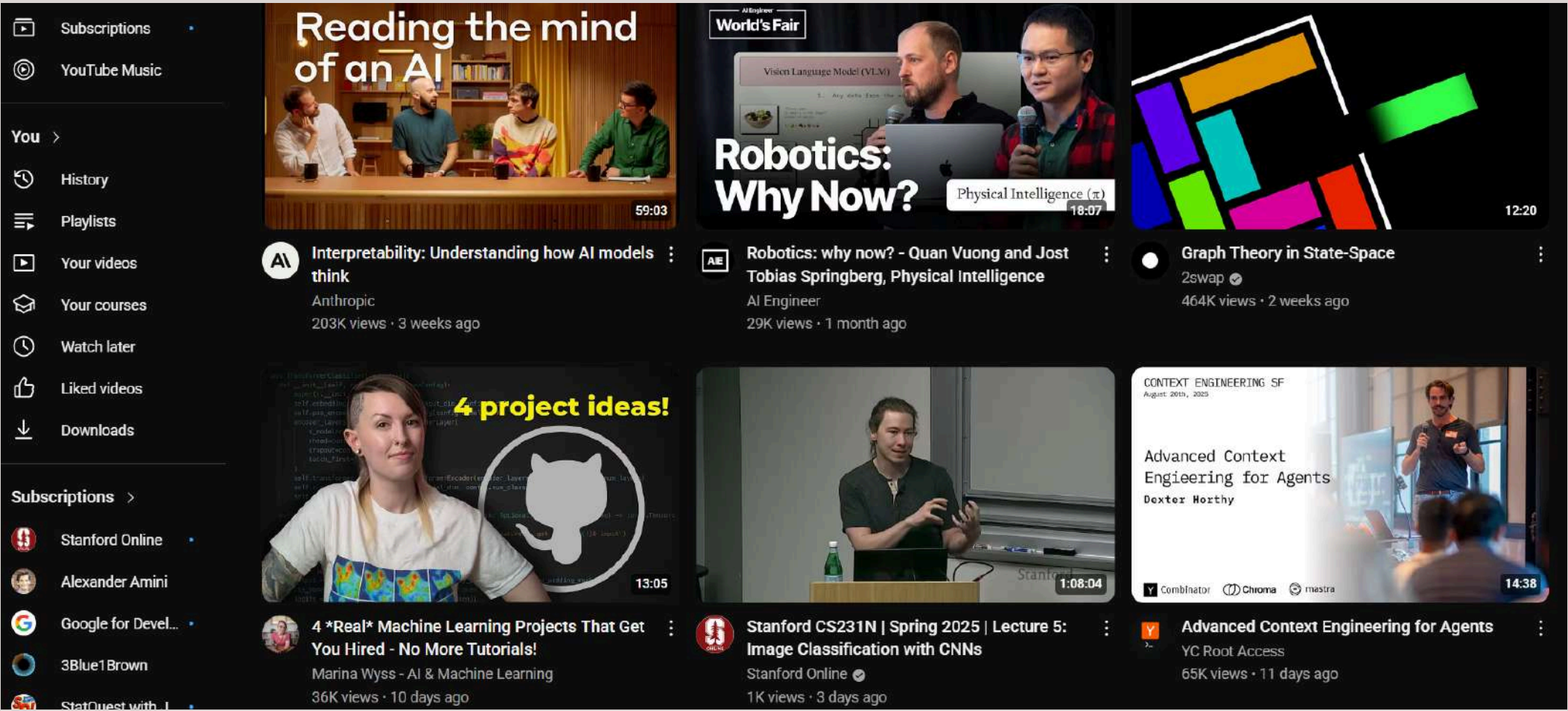
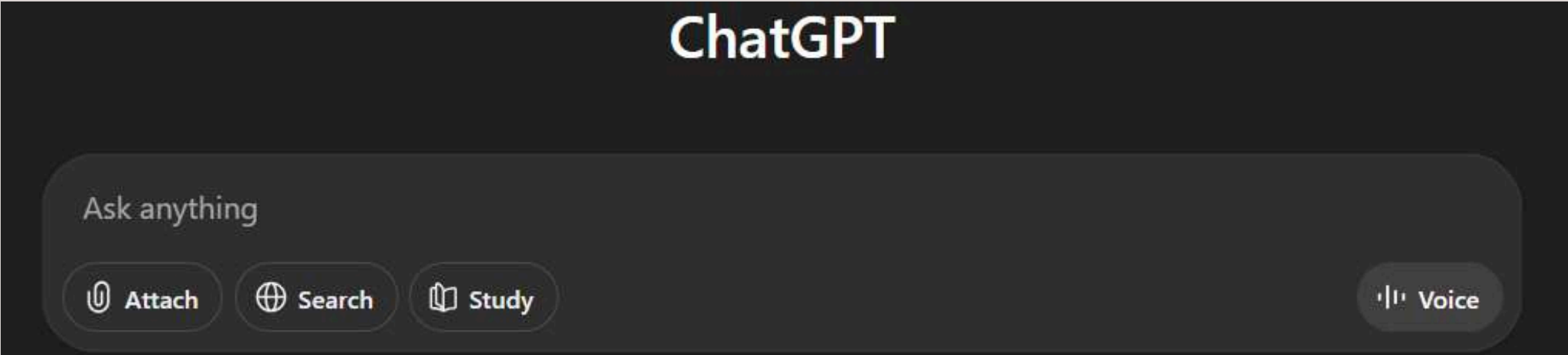


Understand
the core
concepts

Know
the general
workflow

Recognize
the tools

Real-world applications



Is there anyone who has **never created**
a Machine learning model?



Very simple machine learning model
you already created ...

**Harga 4 kg gula dan 10 kg beras adalah
Rp 148.000,00 sedangkan harga 6 kg
gula dan 20 kg beras adalah
Rp 272.000,00. Berapa harga 1 kg gula
dan 5 kg beras ?**

Source: [youtube.com](https://www.youtube.com)

Machine Learning (ML) *Definition*



Arthur Samuel

Field of study that gives computers the **ability to learn without being explicitly programmed**

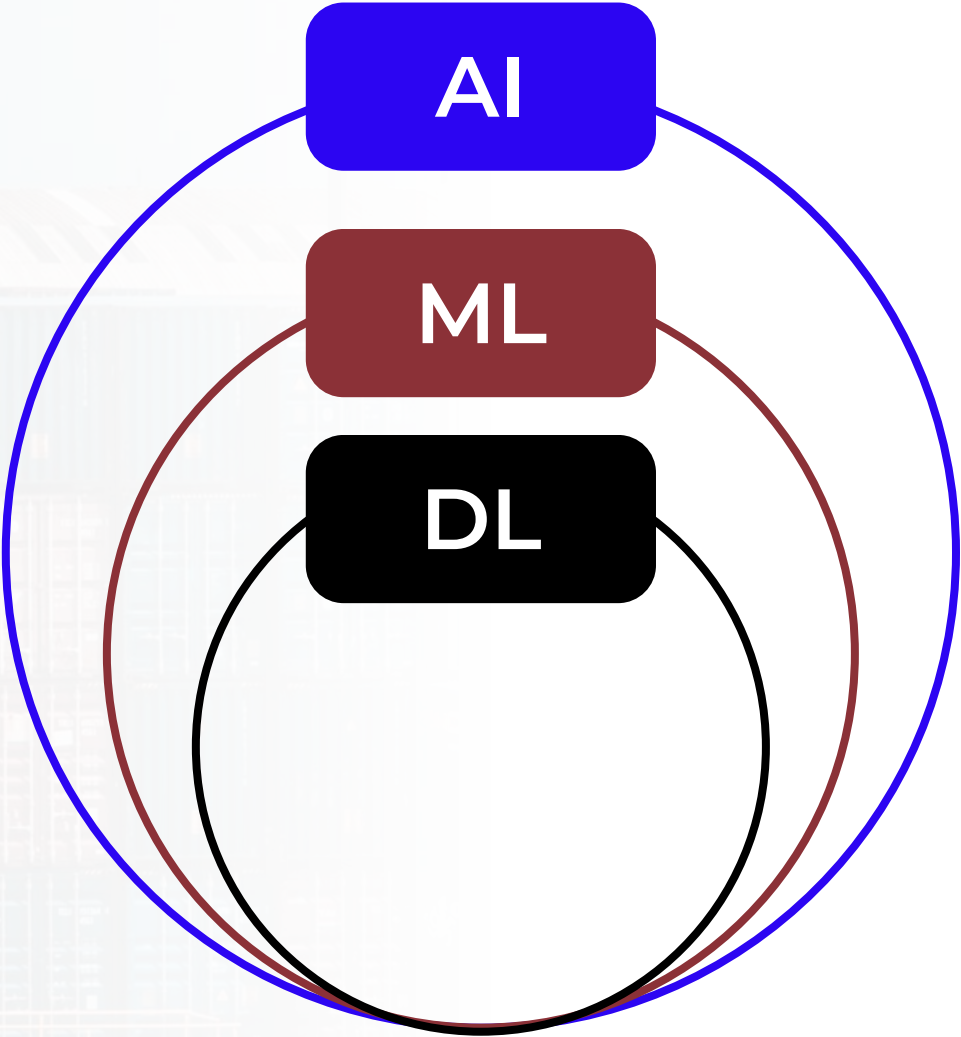


Tom Mitchell

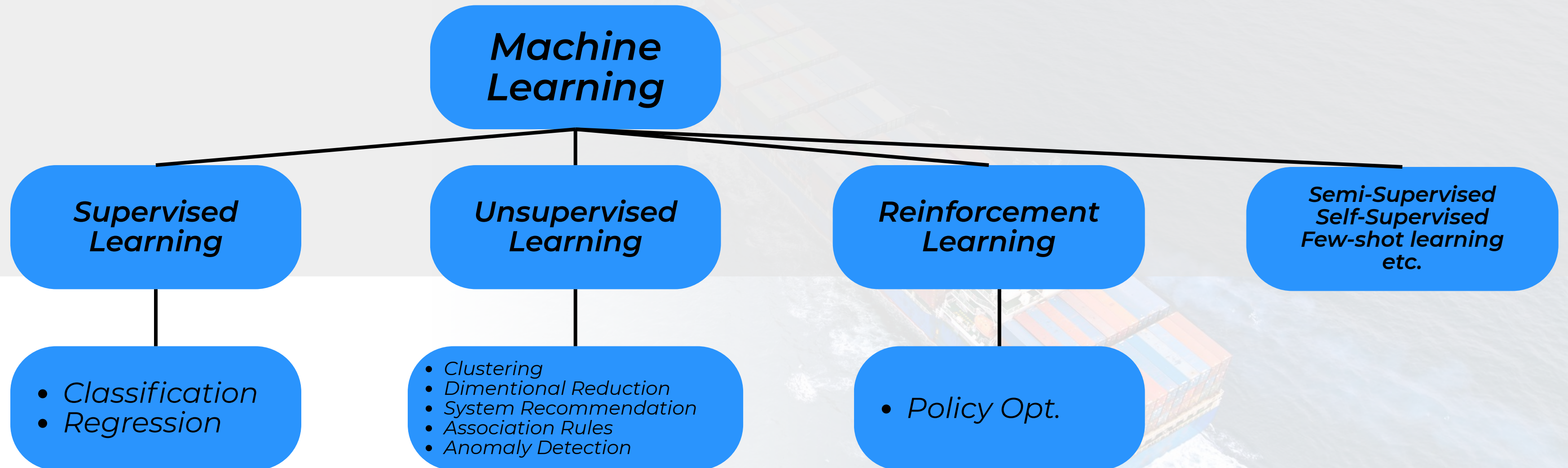
Machine Learning is the study of computer algorithms that **improve automatically through experience**

AI vs ML vs DL

AI	ML	DL
• Broad: create human-like intelligent systems. • Uses rules, search, planning, knowledge, or learning (not only data-driven methods.) • Best when explicit rules or reasoning are required.	• Subset of AI: learn patterns from data. • Typical methods: regression, trees, clustering, etc. • Use for structured/labeled predictive tasks.	• Subset of ML • Learns features end-to-end from raw images/audio/text. (unstructured data). • Needs large data & heavy compute; less interpretable.



Machine Learning Types



$X + y \rightarrow$ Supervised Learning $\rightarrow f(x)$

$X \rightarrow$ Unsupervised Learning $\rightarrow g(X)$ / pattern / structure / representation

Use Case Machine Learning

Regression:

- Forecast energy consumption
- Predict overall energy demand
- Rainfall forecasting

1**2**

Classification:

- Fraud detection
- Defect detection
- Spam filtering

Dimensional Reduction:

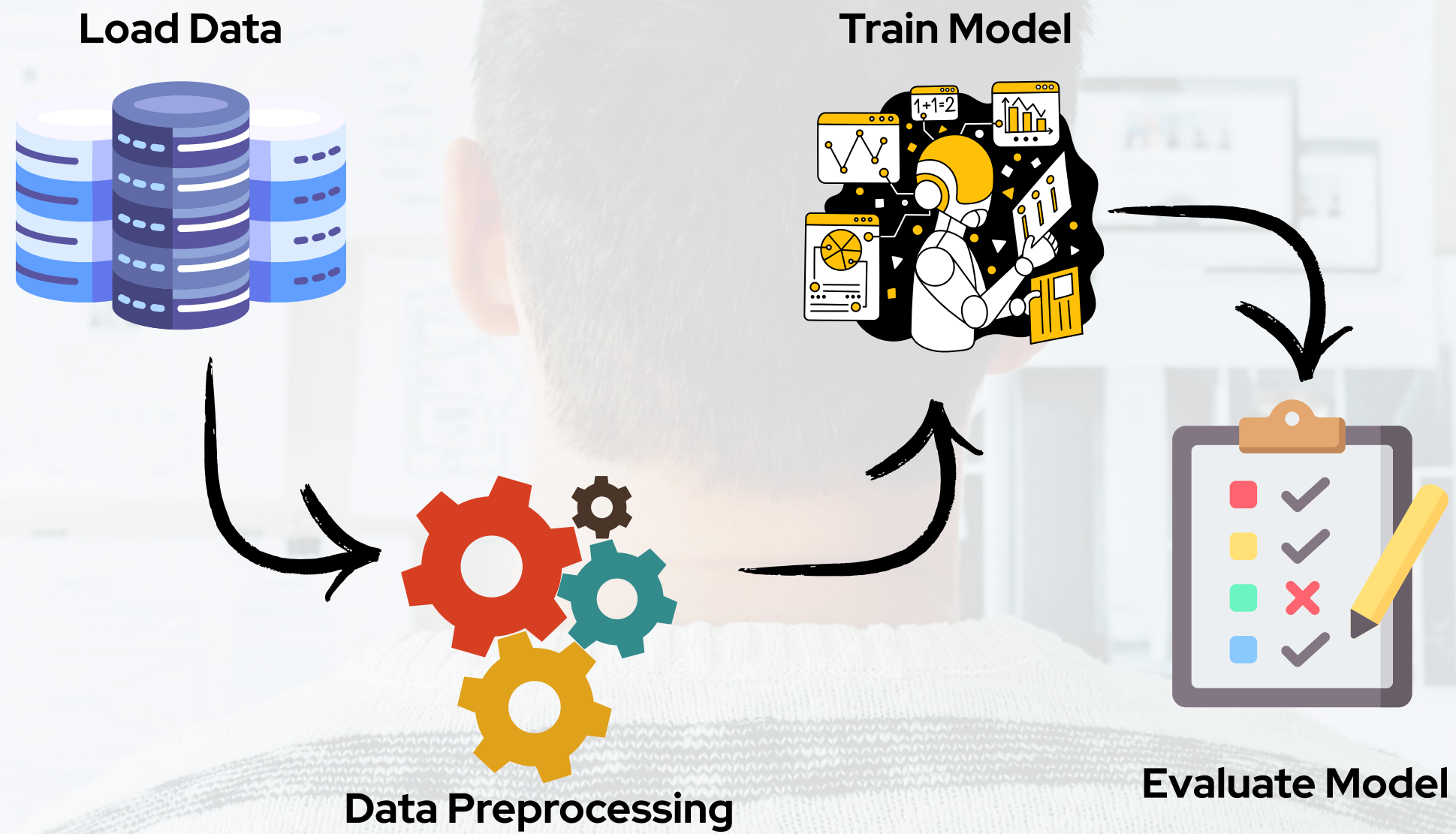
- As a pre-processing part (reduce memory consumption, speed up process)
- Manufacturing process monitoring (Viz High Dim. Data)

3**4**

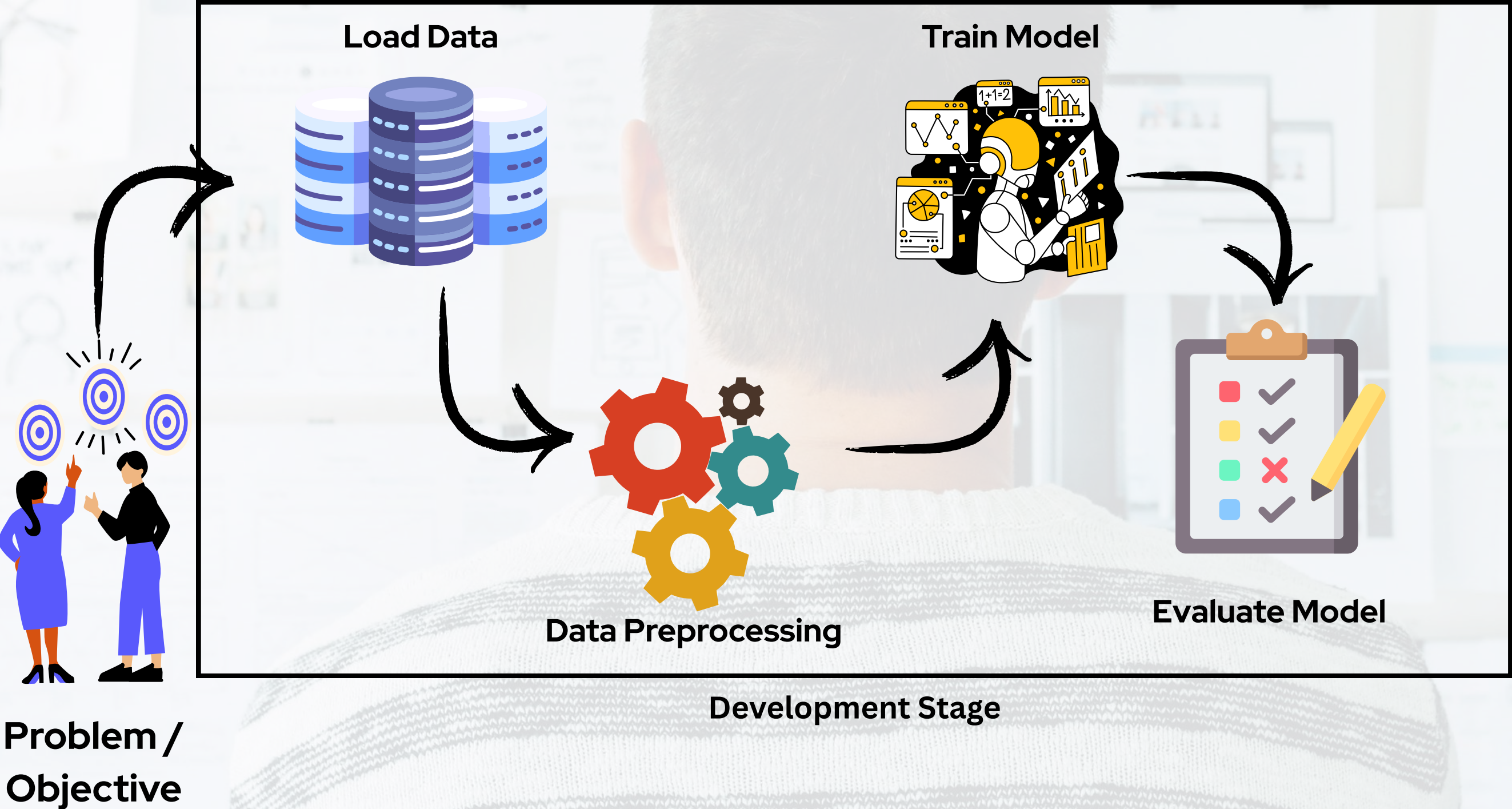
Clustering:

- Clustering suppliers by performance
- Customer segmentation
- Anomaly / Fault Detection

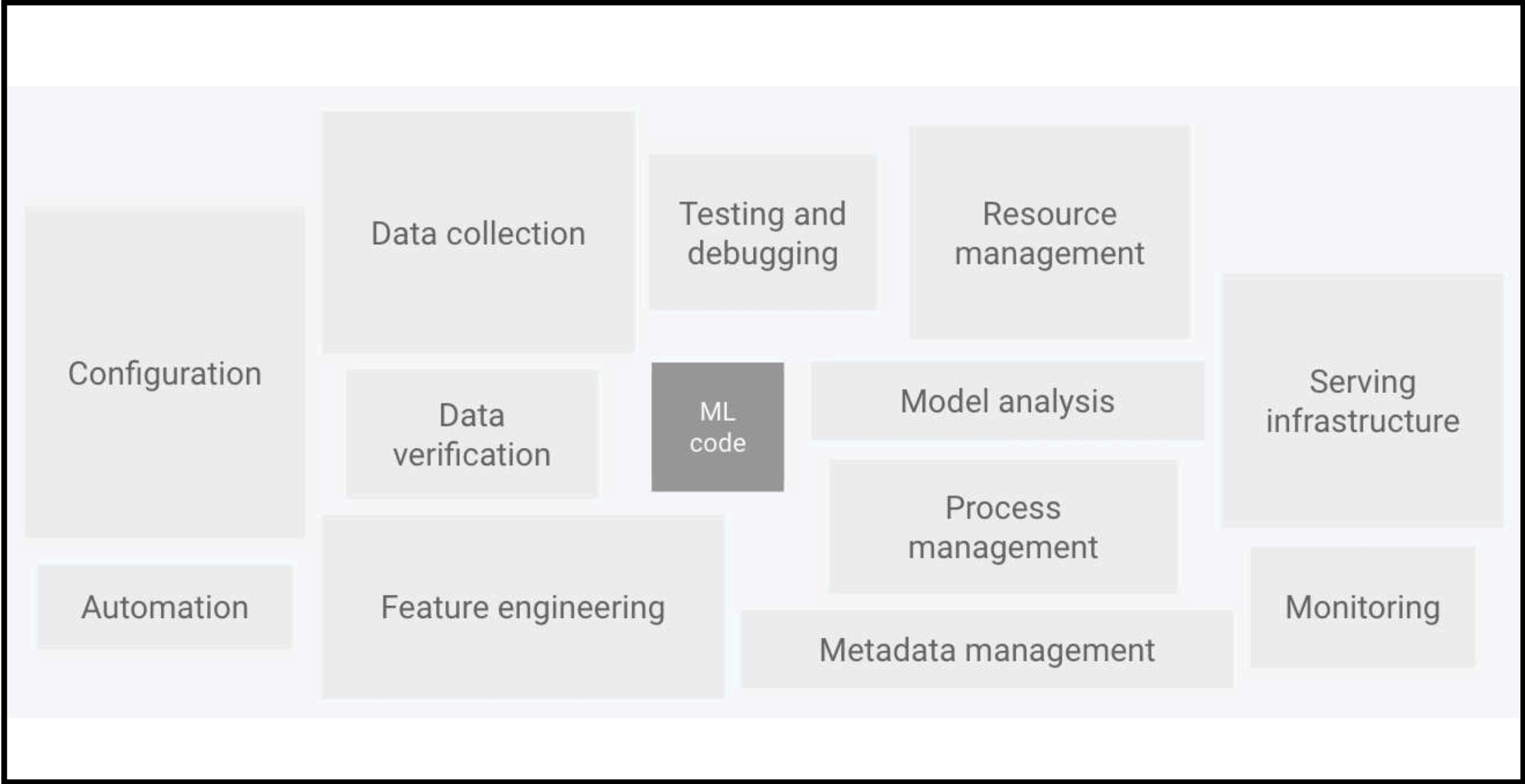
Workflow (Simplified)



Workflow (Simplified)



Deploying a real-world model involves more steps...



Production Stage



Programming Language

Python is the most widely used programming language for machine learning. Some reasons for its popularity include:

- Simple and readable syntax
- Highly versatile
- Widely adopted across industries
- Extensive libraries and frameworks for ML

However, other programming languages can also be used for machine learning, depending on the use case.



```
This example of
Single::ToString( ),
Single::ToString( String* ),
Single::ToString( IFormatProvider* ), and
Single::ToString( String*, IFormatProvider* )
generates the following output when run in the [en-US] culture.
A Single number is formatted with various combinations of format
strings and IFormatProvider.

IFormatProvider is not used; the default culture is [en-US]:
No format string:          11876.54
'N5' format string:       11,876.54000
'E' format string:       1.187654E+004
'E5' format string:       1.18765E+004

A CultureInfo object for [nl-NL] is used for the IFormatProvider:
No format string:          11876.54
'N5' format string:       11.876.54000
'E' format string:       1.187654E+004

A NumberFormatInfo object with digit group size = 2 and
digit separator = ',' is used for the IFormatProvider:
'N' format string:       1_18_76.54
'E' format string:       1.187654E+004
Press any key to continue . . . -
```



Tools



Code Editor

Writing, debugging, & managing code efficiently



Version Control System

experiment tracking & collaboration



Server

Provide the computing power and managed services to train and deploy models.



Container

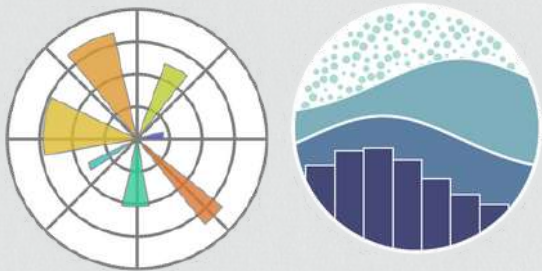
reproducible deployment.



Useful Lib.



Data Processing



Visualization



Deployment



Modeling



Further reading/resources

Machine Learning Lecturer



Local

Course:

- “Pengantar Pembelajaran Mesin”
- “Pengantar Sains Data”
- “Penambangan Data”
- “Kecerdasan Tiruan”
- etc.



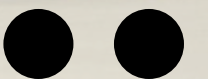
World / Top Univ

Stanford

- [CS229: Machine Learning](#)
- [CS239S: ML System Design](#)
- [CS231n: Deep Learning for Computer Vision](#)

MIT

- [MIT 6.S191: Deep Learning](#)



Let's
discuss!

